Holography in the lowest dimensions: at $d = 0$ it is the wave-function identity, at $d = 1$ it is the Schwarzian

A d-dimensional boundary theory is supposed to be dual to gravity in $d+1$ dimensions. For the two lowest cases the content is very different. For a **zero-dimensional boundary**, the bulk is one-dimensional, where there is no curvature and general covariance reduces to reparametrization of a worldline, so the bulk is the first-quantized particle, classified by its number of endpoints: one point gives the ground-state wave function, two give the propagator, none gives the partition function, and more than two give the Feynman graphs of the target-space theory. The duality at two points is the exact identity

$$|\Omega(x)|^2\,dx \;=\; \text{the } \tau = 0 \text{ marginal of the one-dimensional Euclidean path integral,}$$

the Feynman–Kac form of the ground state; the segment read as a state on its two endpoints is the thermofield double at $\beta = 2T$, and the gap is the rate at which the two ends disentangle as the bulk lengthens. That is, in one dimension lower, the identity used in the transfer note for Yang–Mills, where $|\Omega(U)|^2 d\mu$ is the time-slice marginal of the four-dimensional Wilson measure; the emergent dimension is Euclidean time. For a **one-dimensional boundary**, quantum mechanics, the bulk is two-dimensional dilaton gravity on nearly-AdS_2, the Jackiw–Teitelboim theory, and its dual is the Sachdev–Ye–Kitaev model. Pure two-dimensional gravity has no equations of motion, exact AdS_2 with finite energy fragments, and the surviving dynamics is a single boundary mode, the reparametrization of boundary time with the Schwarzian action $S = -C\int d\tau\,\{f,\tau\}$. This mode gives the near-extremal black hole entropy, saturates the chaos bound with Lyapunov exponent $\lambda_L = 2\pi k_B T/\hbar$, and has a **gapless** density of states $\rho(E) \propto \sinh\left(2\pi\sqrt{2CE}\right)$ with a square-root edge. The naive counting fails at $d = 0$ because the radial direction of anti-de Sitter space is the renormalization scale and a zero-dimensional theory has no renormalization group; for a zero-dimensional scalar integral the bulk is a bare graph, and for a matrix integral the indices thicken the graph into a surface, which is how the double-scaled matrix integrals acquire a two-dimensional bulk. Constants explicit; references at metadata level; nothing promoted.

1. What gravity has to work with in low dimensions

In D spacetime dimensions a massless graviton carries $D(D-3)/2$ polarizations: two in $D = 4$, none in $D = 3$. In $D = 2$ the Einstein tensor vanishes identically and $\sqrt{g}\,R$ is the Euler density, so the Einstein–Hilbert action is a topological number and pure gravity has no field equations; a dilaton ϕ must be added to obtain any dynamics (Teitelboim, Phys. Lett. B 126 (1983) 41; Jackiw, Nucl. Phys. B 252 (1985) 343). In $D = 1$ there is no curvature at all: a one-manifold is flat, and the metric is a single function, the einbein $e(\tau)$, with reparametrization as the only gauge symmetry.

So in both low cases the bulk has no local gravitational degrees of freedom, and all the physics sits at the boundary or in the topology.

2. Zero-dimensional boundary, one-dimensional bulk

The bulk theory is the relativistic particle,

$$S = \frac{1}{2}\int d\tau\Big[\frac{1}{e}\,\dot{x}^2 + e\,m^2c^2\Big],$$

whose constraint is the mass shell $p^2 = m^2c^2$ and whose only "gravitational" content is the invariance under $\tau \to f(\tau)$. A connected one-manifold with boundary is a segment or a half-line, a connected one-manifold without boundary is a circle, and a connected one-dimensional

space with more than two ends is a graph with junctions. The boundary is a finite set of points, and **the number of points decides what the bulk computes.**

bulk one-manifold	boundary points	the bulk path integral equals
half-line $[0,\infty)$	1	the ground-state wave function $\Omega(x)$
segment $[0,T]$	2	the propagator $K_T(x,y)=\langle x\|e^{-TH/\hbar}\|y\rangle$
circle of length β	0	the partition function $\operatorname{Tr} e^{-\beta H/\hbar}$
k disjoint segments	$2k$	the product of k propagators
graph with k ends	k	a Feynman diagram of the target-space theory

One point: the wave function. Fix one end at x and let the other run to infinity with any boundary state χ of nonzero overlap with Ω; then

$$\lim_{T\to\infty} e^{TE_0/\hbar}\langle x|e^{-TH/\hbar}|\chi\rangle = \Omega(x)\,\langle\Omega,\chi\rangle,$$

so a single boundary point carries the ground-state wave function, and the half-line is its Euclidean history. This is the no-boundary prescription of Hartle and Hawking (Phys. Rev. D 28 (1983) 2960) in its one-dimensional form: the wave function is the path integral over half-geometries ending on the boundary.

Two points: the propagator, and the gap. With both ends fixed the bulk computes $K_T(x,y)$, and the spectral decomposition gives

$$K_T(x,y) = e^{-TE_0/\hbar}\,\Omega(x)\Omega(y)\Big[1+O\big(e^{-T\Delta/\hbar}\big)\Big], \qquad \Delta = E_1 - E_0.$$

The leading term is the product of two half-lines, one per boundary point; **the gap is the decay rate, in the bulk length, of the connected part**. In the language of ensembles, a connected bulk joining two boundary components is a wormhole, and the statement that its contribution factorizes up to $e^{-T\Delta/\hbar}$ says that in one dimension the wormhole's failure to factorize is exactly the gap. For every bounded F, Feynman–Kac then gives

$$\langle\Omega, F\Omega\rangle = \lim_{T\to\infty} \frac{\int \mathcal{D}x\; F(x(0))\; e^{-S_E[x]/\hbar}}{\int \mathcal{D}x\; e^{-S_E[x]/\hbar}}, \qquad \frac{S_E}{\hbar} = \frac{1}{\hbar}\int_{-T}^{T} d\tau\Big[\frac{m}{2}\dot{x}^2 + W(x)\Big],$$

so $|\Omega|^2 dx$ is the $\tau=0$ marginal of the segment's measure. That is Proposition 1 of the transfer note, one dimension lower: there the segment is the Euclidean slab $[-T,T]\times\mathbb{T}^3$, its two "points" are the two time slices, and the Yang–Mills gap is the decay of the connected slice-to-slice correlator with T. In both cases the reason to move to the bulk is the same: the bulk weight $e^{-S_E/\hbar}$ factorizes over time steps and separates configurations locally, while the boundary eigenvalue equation $-\frac{\hbar^2}{2m}\Omega'' + (W-E_0)\Omega = 0$ compares the potential with the total energy. The Agmon estimate of the Agmon note is the boundary-side statement; the Feynman–Kac marginal is the bulk-side one.

Cutting the segment. Inserting a time point $\tau\in(0,T)$ and integrating over the position there,

$$K_T(x,y) = \int dz\; K_{T-\tau}(x,z)\,K_\tau(z,y),$$

is the semigroup property $e^{-TH/\hbar} = e^{-(T-\tau)H/\hbar}e^{-\tau H/\hbar}$ with the resolution of the identity $\int dz\,|z\rangle\langle z|$ at the cut. The amplitude is unchanged, so a cut decomposes the wormhole into two shorter ones glued along the cut, and inserts nothing. Iterating the cuts at spacing a_t is the

transfer-matrix form $e^{-TH/\hbar} = \mathcal{T}^{T/a_t}$, and the cut at $\tau = 0$ of a slab $[-T, T]$ is exactly how the transfer note exhibits $|\Omega|^2$ as a marginal.

What the segment is, read as a state rather than as an amplitude, is sharper. Regard $K_T(x, y)$ as the wave function of a state on the two boundary points, an element of $\mathcal{H} \otimes \mathcal{H}$. Its Schmidt decomposition is the spectral one,

$$K_T = \sum_n e^{-TE_n/\hbar}\ \psi_n \otimes \overline{\psi_n},$$

which, after normalization and identification of the second factor with the conjugate Hilbert space, is the thermofield double at physical inverse temperature $\beta_E = 2T/\hbar$. Assume a simple ground state, an isolated first excitation Δ of finite degeneracy d_1, and a heat trace finite at some positive inverse temperature. The two ends are entangled with Schmidt weights $p_n \propto e^{-2TE_n/\hbar}$, so their entropy equals the thermal entropy and has large-T asymptotics

$$S_{\text{ent}}(T) = d_1 \left(1 + \frac{2T\Delta}{\hbar}\right) e^{-2T\Delta/\hbar} + o(e^{-2T\Delta/\hbar}).$$

After subtracting E_0, put $x = 2T\Delta/\hbar$. The partition function is $1 + d_1 e^{-x}$ plus higher-energy terms, and $S = \log Z + \beta_E \langle H - E_0 \rangle$ gives the coefficient. The d_1 states are separate Schmidt weights; there is no $\log(1/d_1)$ term. **The gap is the rate at which the two ends of the segment disentangle as the bulk lengthens.** Factorization of the wormhole at large T and disentanglement of the thermofield double are the same statement. For the Yang–Mills slab of the transfer note, the wave functional $\Psi(U_{-T}, U_T) = \langle U_T | e^{-2TH/\hbar} | U_{-T} \rangle$ of the two boundary slices is the thermofield double of the lattice Hamiltonian at $\beta_E = 4T/\hbar$. Its entropy uses the same formula with T replaced by $2T$. The semigroup duration, physical inverse temperature and energy gap retain these conversion factors.

The ensemble wormholes of Section 3, a cylinder joining two circles, have no one-dimensional counterpart: a connected one-manifold with two boundary components is a segment, whose components are points. Their one-dimensional shadow is the segment, which needs no ensemble.

No points: the spectrum. The circle has empty boundary and computes $\operatorname{Tr} e^{-\beta H/\hbar} = \sum_n e^{-\beta E_n/\hbar}$, which at large β reads $e^{-\beta E_0/\hbar}(1 + e^{-\beta\Delta/\hbar} + \dots)$. A bulk with no boundary at all still knows the gap, through its length.

More than two points: graphs and Feynman diagrams. Disjoint segments factorize. A connected bulk with $k > 2$ ends is a graph, and pure one-dimensional gravity supplies no rule at the junctions; the junction rule is extra data, and once it is given the sum over graphs with edge lengths integrated is the Schwinger proper-time form of the Feynman expansion (Schwinger, Phys. Rev. 82 (1951) 664): each edge is a segment computing a propagator, each junction is a vertex of the target-space theory, and the einbein moduli are the Schwinger parameters. The "boundary theory" is then the field theory on the target space of the particle, in as many dimensions as the target has.

The zero-dimensional field theory proper. Take the target space to be a point. The propagator is a number, the edges have no moduli, and the bulk is a bare graph: the Feynman expansion of an ordinary integral $\int d\phi\, e^{-V(\phi)/\hbar}$ is a sum over graphs weighted by symmetry factors and couplings, with no geometry left on the edges. This is the exact content of "holography for a zero-dimensional theory": **the bulk of a zero-dimensional scalar or vector integral is a one-dimensional graph with no metric.** When the variable is an $N \times N$ matrix, each graph acquires an index structure that thickens it into a ribbon graph, that is, a two-dimensional surface with a genus expansion in $1/N$ ('t Hooft, Nucl. Phys. B 72 (1974) 461), and that is the mechanism by which the double-scaled matrix integrals of Section 3 acquire a two-dimensional bulk. The extra dimension comes from the matrix indices, and the count "bulk = boundary + 1" is restored by the thickening rather than by any property of the point.

3. One-dimensional boundary, two-dimensional bulk

Here the duality has content. The bulk is Jackiw–Teitelboim gravity,

$$\frac{S}{\hbar} = -\frac{1}{16\pi G}\Big[\int d^2x\sqrt{g}\,\phi\Big(R+\frac{2}{L^2}\Big) + 2\int_\partial \sqrt{h}\,\phi_b\,K\Big],$$

in which the dilaton fixes $R=-2/L^2$ and the geometry is AdS_2 exactly. Exact AdS_2 cannot support finite-energy excitations: the backreaction of any finite energy destroys the asymptotics (Maldacena, Michelson and Strominger, JHEP 1999 (02) 011), so the correspondence is formulated on *nearly-AdS_2*, where the dilaton grows toward the boundary and breaks the conformal symmetry (Almheiri and Polchinski, JHEP 2015 (11) 014; Maldacena, Stanford and Yang, PTEP 2016, 12C104).

The Schwarzian mode. With the bulk geometry rigid, the only dynamical variable is the shape of the boundary curve, a reparametrization $f(\tau)$ of boundary time, and the action reduces to

$$\frac{S}{\hbar} = -C\int d\tau\ \{f,\tau\}, \qquad \{f,\tau\} = \frac{f'''}{f'} - \frac{3}{2}\Big(\frac{f''}{f'}\Big)^2,$$

with $C\propto \phi_b/G$. Its consequences: the near-extremal black hole entropy $S = S_0 + 2\pi^2 C/\beta\,(\cdot k_B)$ to leading order in $1/\beta$; the out-of-time-order correlator growing at the maximal rate $\lambda_L = 2\pi k_B T/\hbar$; and the density of states

$$\rho(E)\ \propto\ \sinh\Big(2\pi\sqrt{2CE}\Big),$$

which vanishes at $E=0$ like $\sqrt{E}$ and has no gap.

The boundary dual. The Sachdev–Ye–Kitaev model, N Majorana fermions with random all-to-all q-body couplings (Sachdev and Ye, Phys. Rev. Lett. 70 (1993) 3339; Kitaev, KITP talks 2015; Maldacena and Stanford, Phys. Rev. D 94 (2016) 106002), has at large N and low energy exactly the Schwarzian as its effective action, the same entropy, the same maximal chaos, and the same gapless spectrum. The matching is at the level of the low-energy effective theory rather than of a single microscopic theory: the SYK couplings are drawn from an ensemble, and the bulk sum over topologies is the dual of the ensemble average (Saad, Shenker and Stanford, arXiv:1903.11115, which shows that JT gravity summed over all genera is a double-scaled matrix integral).

4. Why the naive counting fails at $d=0$

The radial coordinate of AdS_{d+1} is the renormalization scale of the boundary theory; an emergent dimension needs a continuum of scales to emerge from. A zero-dimensional theory is a finite-dimensional integral and has no renormalization group, so the $d=0\to D=1$ reading of the counting has nothing to build the dimension out of, and Section 2 is the whole of what it contains: Euclidean time as the emergent direction, and the Feynman–Kac identity as the dictionary. The dualities with zero-dimensional boundaries that do carry content, the double-scaled matrix integrals of Section 3, have **two-dimensional** bulks. Their extra dimension comes from the matrix indices thickening each graph into a ribbon graph, with the eigenvalue density playing the role of a continuum of scales and the genus expansion supplying the sum over topologies; the boundary object is an ensemble of theories rather than one.

5. Bearing on this programme

Three points.

1. The $d=0$ identity of Section 2 is the tool that the transfer note uses in $3+1$ dimensions. In both cases the reason to move to the bulk is the same: the bulk weight factorizes and

yields local estimates, the boundary eigenvalue equation carries the extensive energy and does not.

2. The $d = 1$ example is a scale generated entirely by a boundary mode, and it is gapless. A Schwarzian-type mechanism therefore produces a scale without producing a gap, which is the opposite of what the Yang–Mills problem needs and a useful reminder that the two are distinct.
3. The holographic setting in which a gap does appear is a bulk whose infrared end caps off: Witten's confining geometry, in which the Euclidean time circle shrinks smoothly to zero size and the bulk spectrum acquires a discrete tower with a gap set by the size of the cap (Witten, Adv. Theor. Math. Phys. 2 (1998) 505). That construction lives at large N and strong 't Hooft coupling, so it gives a picture of how a gap arises in a different regime from the weak-coupling one this programme works in, and no theorem about $SU(3)$.

6. Consequence for STATE

Recorded for orientation. The $d = 0$ case is the Feynman–Kac identity that the programme already uses; the $d = 1$ case is the Schwarzian/SYK system, gapless with $\rho(E) \propto \sinh(2\pi\sqrt{2CE})$; the holographic mechanism for a gap is a capped-off infrared geometry at large N. Nothing in the main line changes.